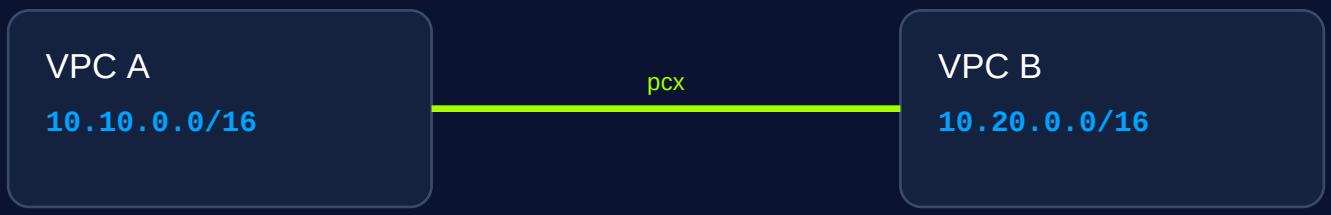
EWB COURSES

BUILD / TEST / EXPLAIN

AWS

VPC Peering

Connect two VPCs.
Prove private connectivity.



Student Project Report

Name:	Dilli Prathap
Batch:	1.1
Region:	Asia Pacific (Mumbai) / ap-south-1
AWS account:	902916048053
Report date:	30 October 2026

1. Project summary and resource inventory

This report documents my build of the EWB AWS VPC Peering project in the Mumbai region (ap-south-1): two non-overlapping VPCs (VPC A 10.10.0.0/16 and VPC B 10.20.0.0/16), public and private subnets with explicit route table associations, an internet gateway on VPC A only, security groups for the application servers, a VPC peering connection that is Active, EC2 instances in VPC A, and the cleanup of all resources afterwards. Every screenshot is labelled with the console page, the checkpoint and what it shows.

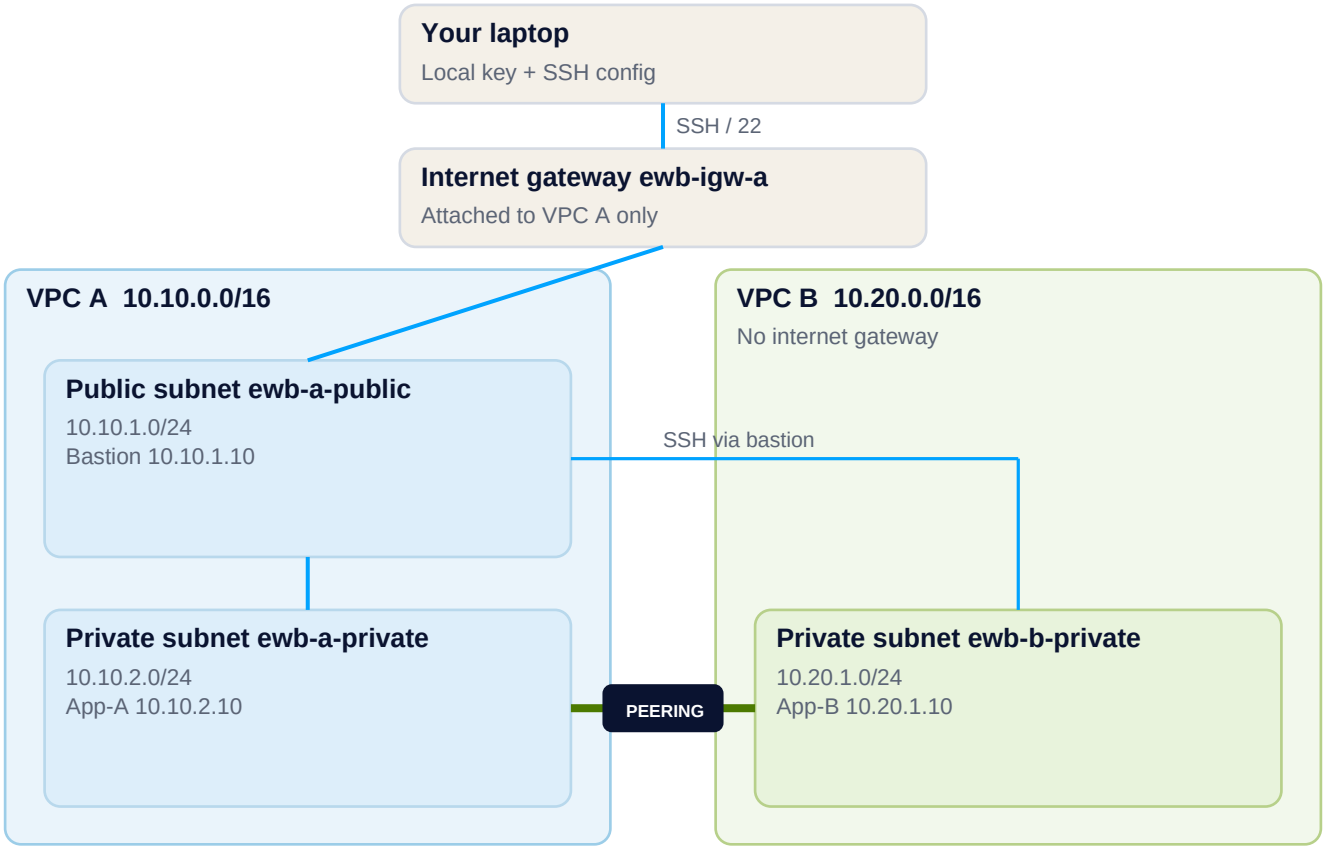
Resource inventory (IDs read from my screenshots)

Resource / Name tag	CIDR / IP	AWS ID	State
ewb-vpc-a	10.10.0.0/16	vpc-0dc3eaaa48f8bd4ba	Available
ewb-a-public	10.10.1.0/24	subnet-063e2fa071b39b60d	Available
ewb-a-private	10.10.2.0/24	subnet-0d9f52d2a3d963220	Available
ewb-igw-a	VPC A only	igw-0ea88add97112536f	Attached
ewb-rt-a-public	0.0.0.0/0 to IGW	rtb-0a0b81ff85754de10	Associated
ewb-rt-a-private	local only	rtb-01f82e7a496913f21	Associated
ewb-vpc-b	10.20.0.0/16	vpc-0a2d862236d5b99f8	Available
ewb-b-private	10.20.1.0/24	subnet-026d9fda11eea3237	Available
ewb-rt-b-private	local only	rtb-05280c9b35580f8ee	Associated
ewb-a-b-peer	VPC A to VPC B	pcx-0ee2feeb51a3dda9a	Active
ewb-sg-app-a	SSH + ICMP echo	sg-062179dbba0becb82	Created
ewb-sg-app-b	SSH + ICMP echo	sg-01dbc66e1827a21a9	Created
ewb-bastion	10.10.1.10 / public 43.205.229.108	i-01a30032838ce244d	Running, 3/3 checks
ewb-app-a	10.10.2.10, no public IP	i-0ab37d64ad8477813	Running, 3/3 checks

Tags applied: Project = EWB-VPC-Peering, Student = Dilli-Prathap (visible on the internet gateway screenshot).

2. Architecture and address plan

Administration path: laptop > bastion > App-A or App-B using SSH ProxyJump. Project traffic: App-A <> App-B over private IPs through the peering connection. Only the bastion has a public IPv4 address. VPC B has no internet gateway.



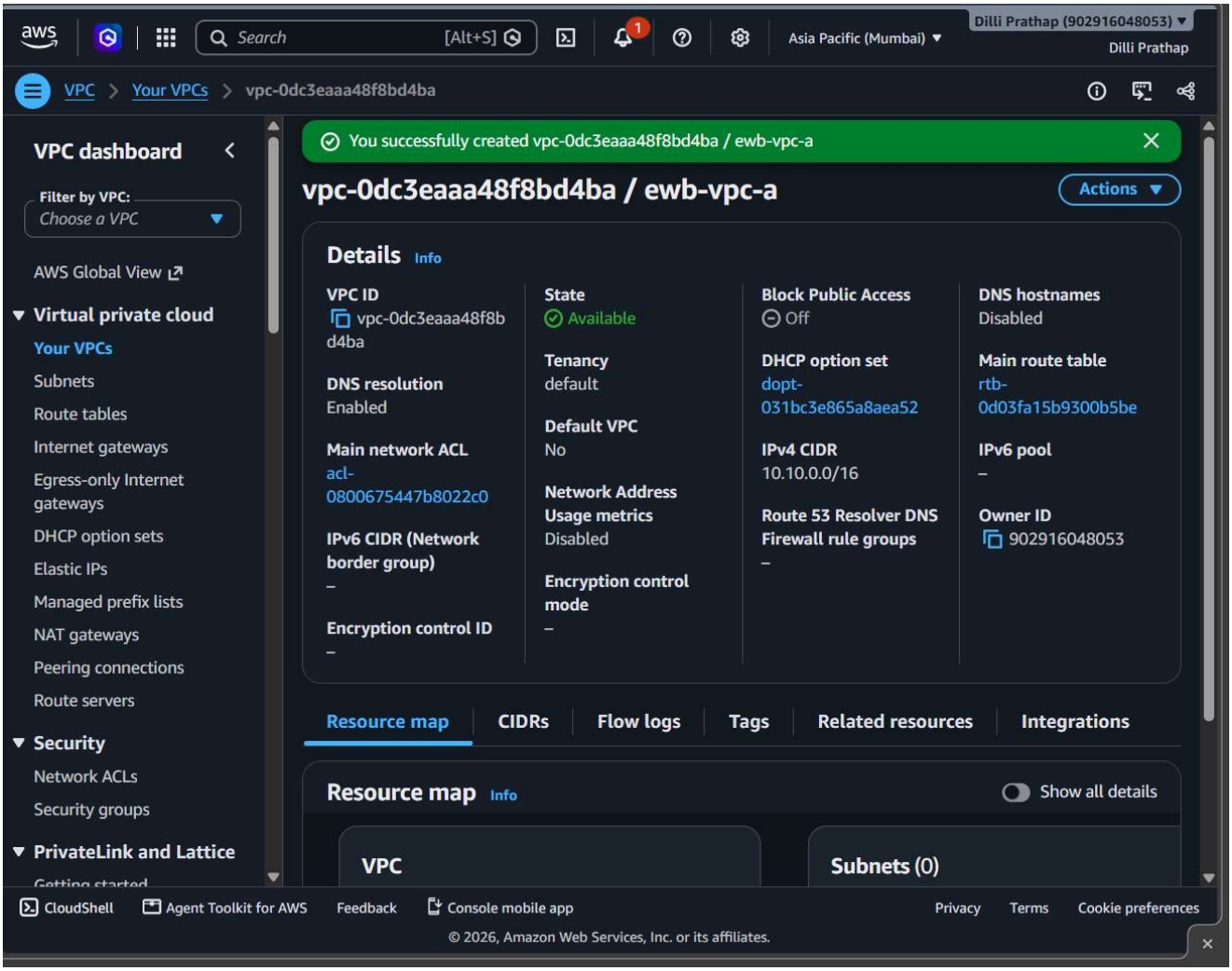
Why these CIDRs? 10.10.0.0/16 and 10.20.0.0/16 do not overlap, and each /24 subnet sits inside its own VPC range. VPC peering cannot be created between VPCs with overlapping CIDRs.

Diagram is conceptual; the console evidence follows.

3. Build evidence - VPC A

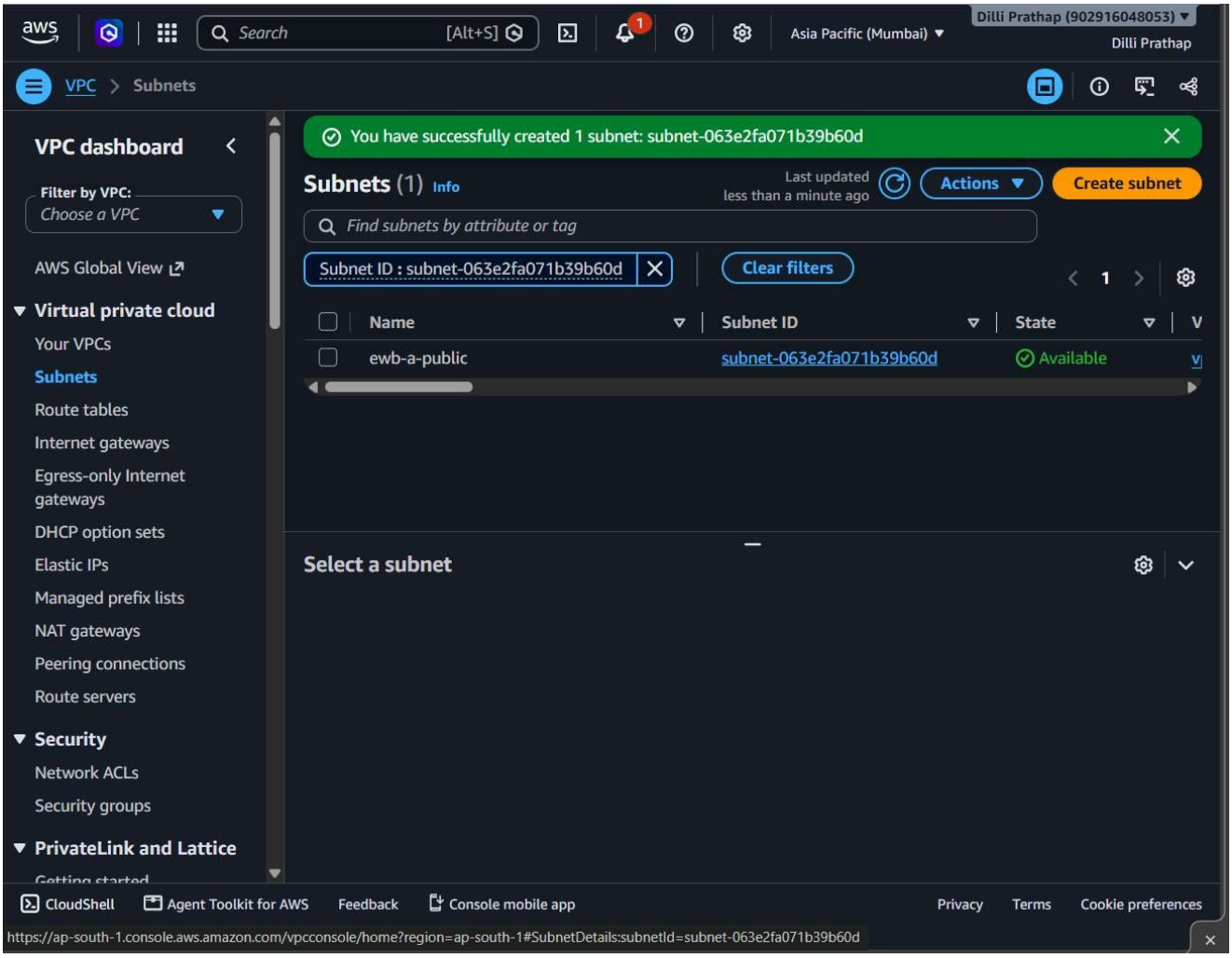
3.1 VPC A created (ewb-vpc-a)

Host / console page	Test / checkpoint	Result
VPC console > Your VPCs	Create VPC	State Available; IPv4 CIDR 10.10.0.0/16; ID vpc-0dc3eaaa48f8bd4ba



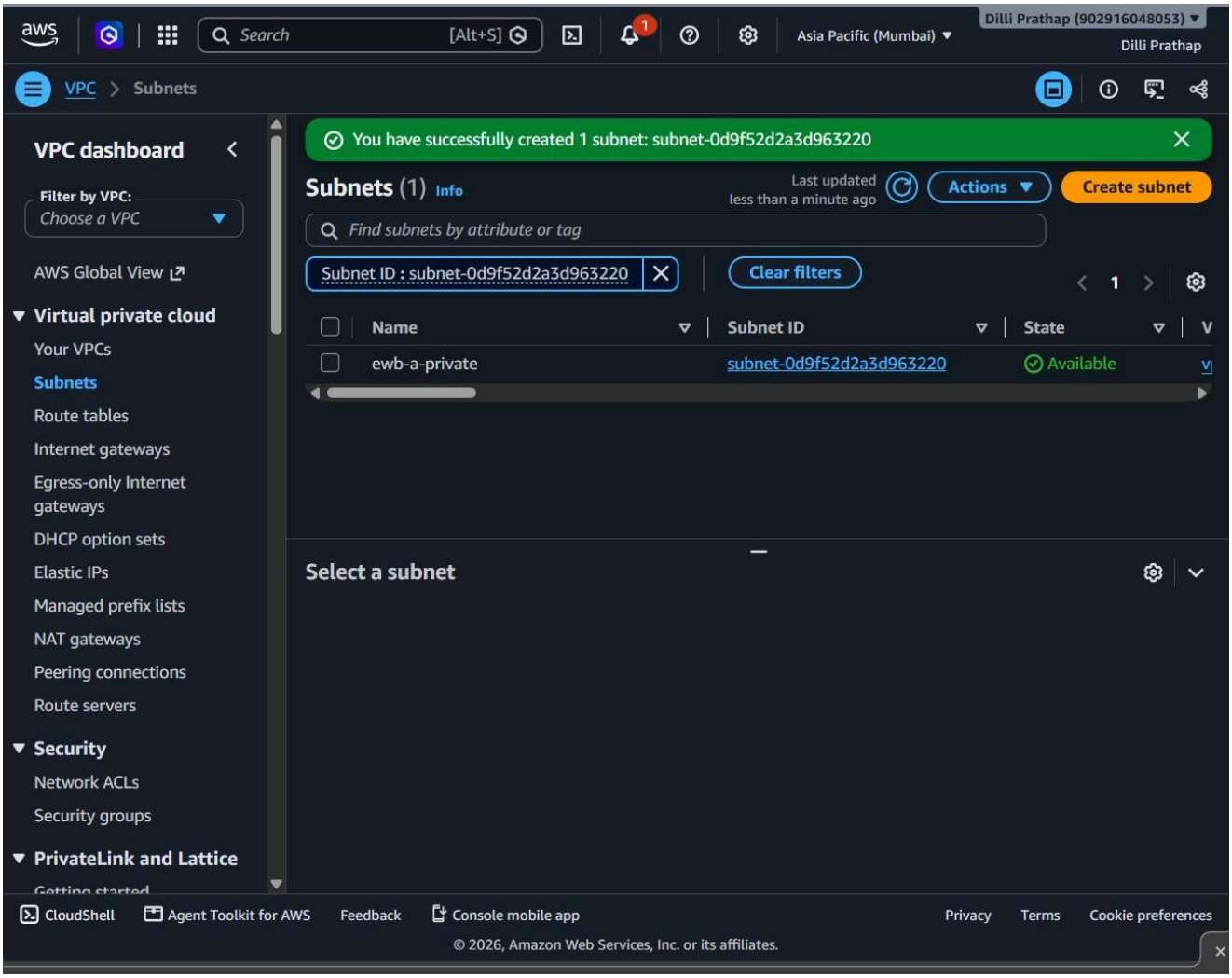
3.2 Public subnet created (ewb-a-public)

Host / console page	Test / checkpoint	Result
VPC console > Subnets	Create subnet	subnet-063e2fa071b39b60d is Available



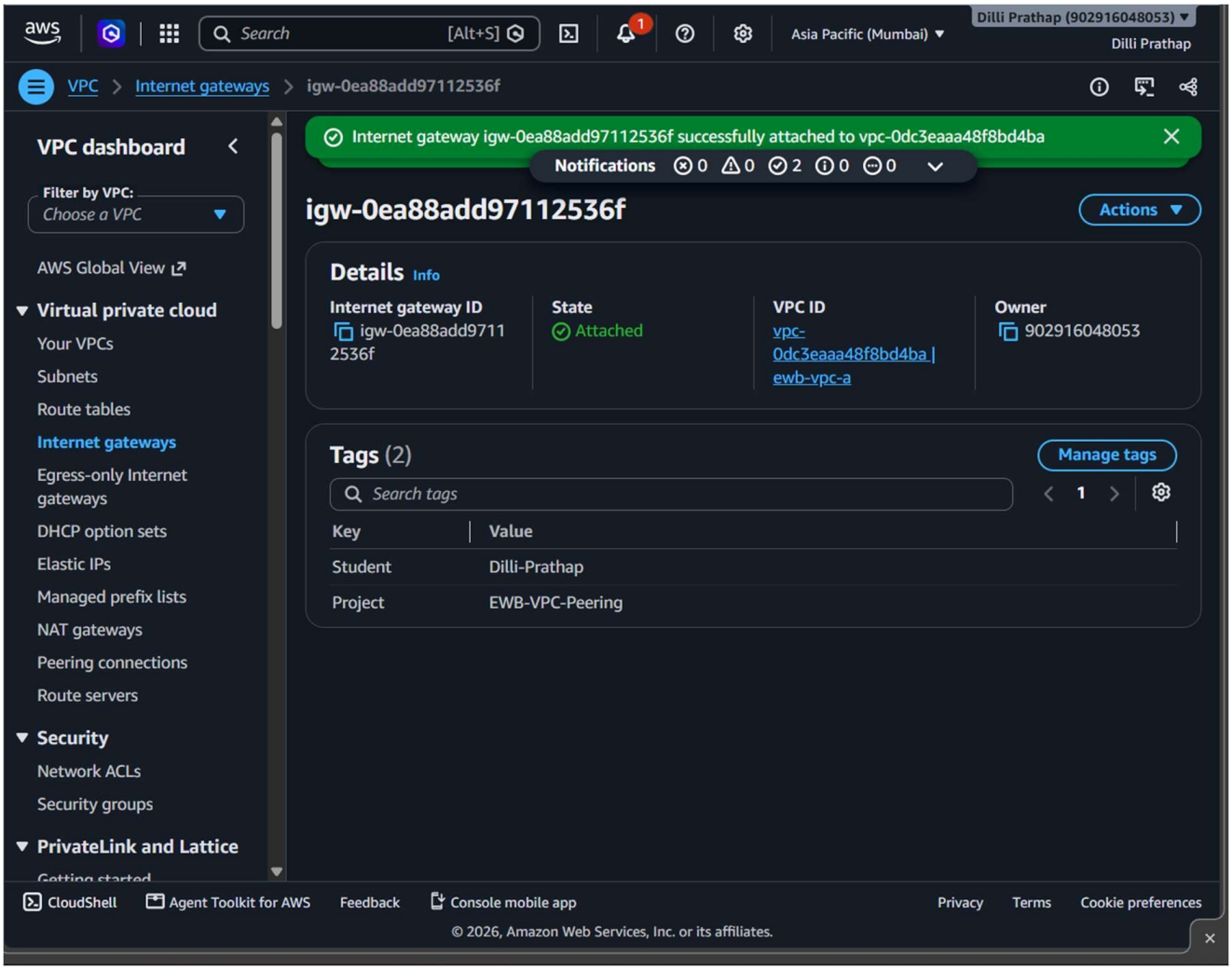
3.3 Private subnet created (ewb-a-private)

Host / console page	Test / checkpoint	Result
VPC console > Subnets	Create subnet	subnet-0d9f52d2a3d963220 is Available



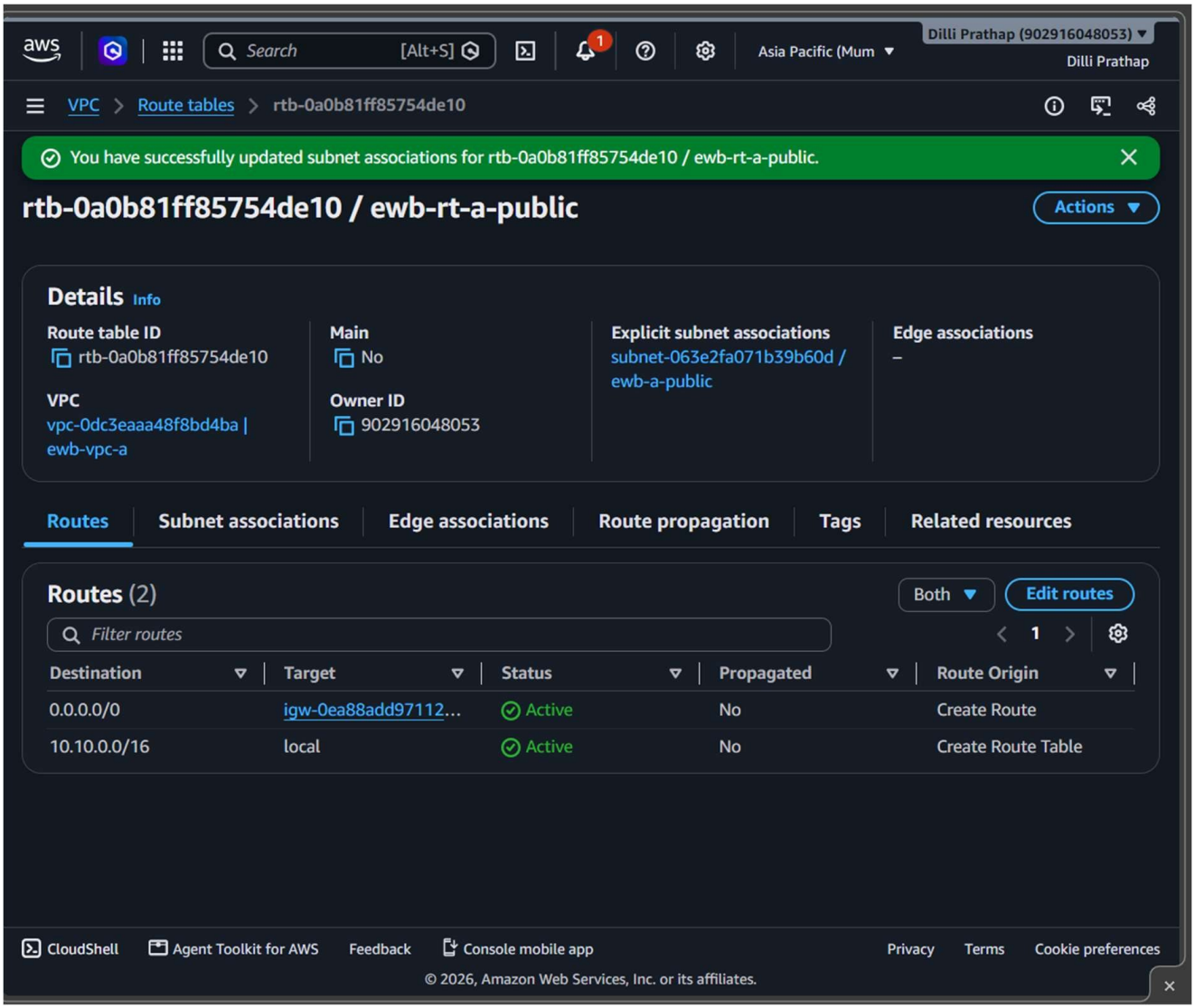
3.4 Internet gateway attached to ewb-vpc-a

Host / console page	Test / checkpoint	Result
VPC console > Internet gateways	Create and attach IGW	igw-0ea88add97112536f State Attached to vpc-0dc3eaaa48f8bd4ba; tags Student and Project present



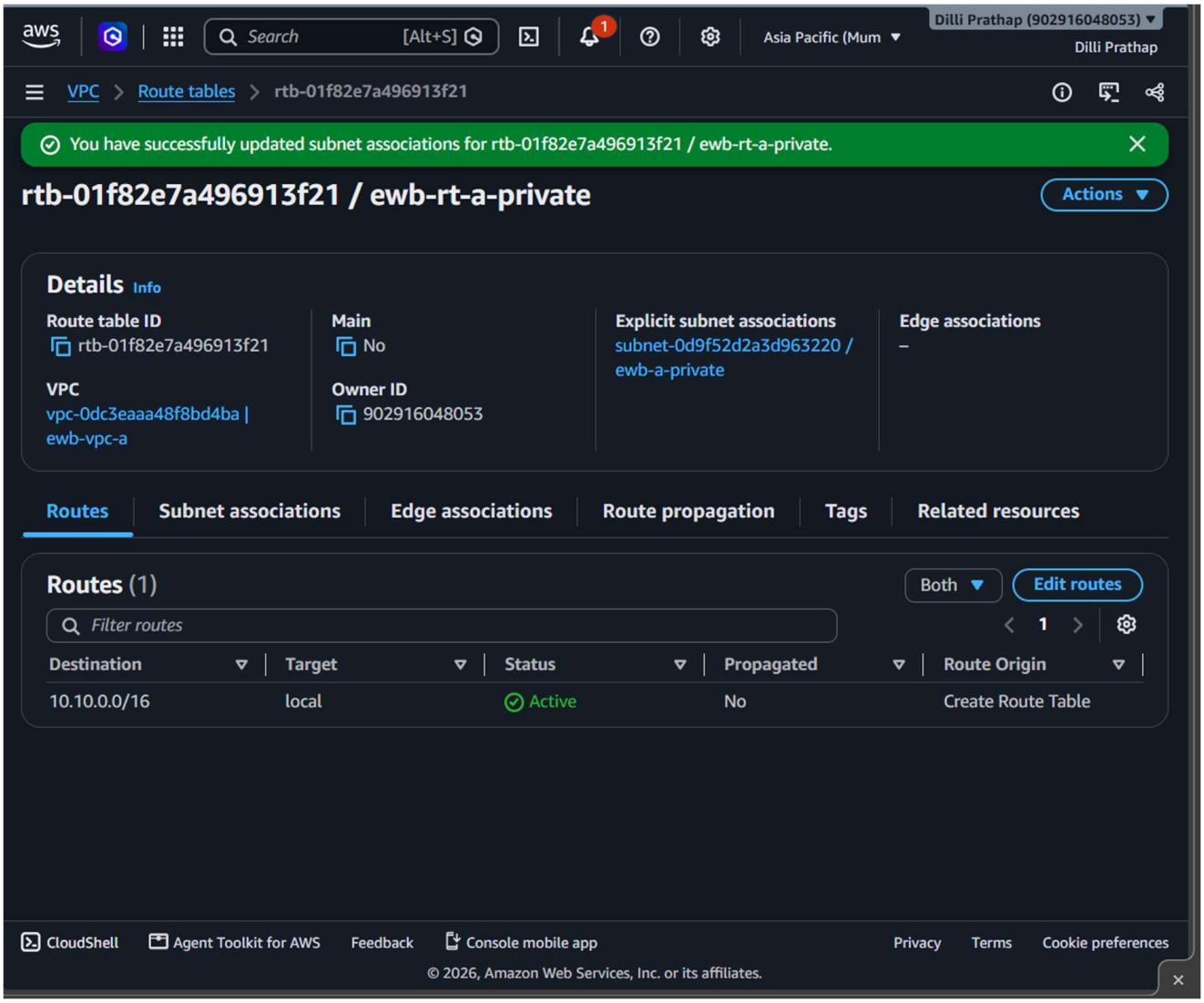
3.5 Public route table (ewb-rt-a-public)

Host / console page	Test / checkpoint	Result
Route tables > ewb-rt-a-public	Association and default route	Associated with ewb-a-public. Routes: 0.0.0.0/0 to the internet gateway, 10.10.0.0/16 local



3.6 Private route table (ewb-rt-a-private)

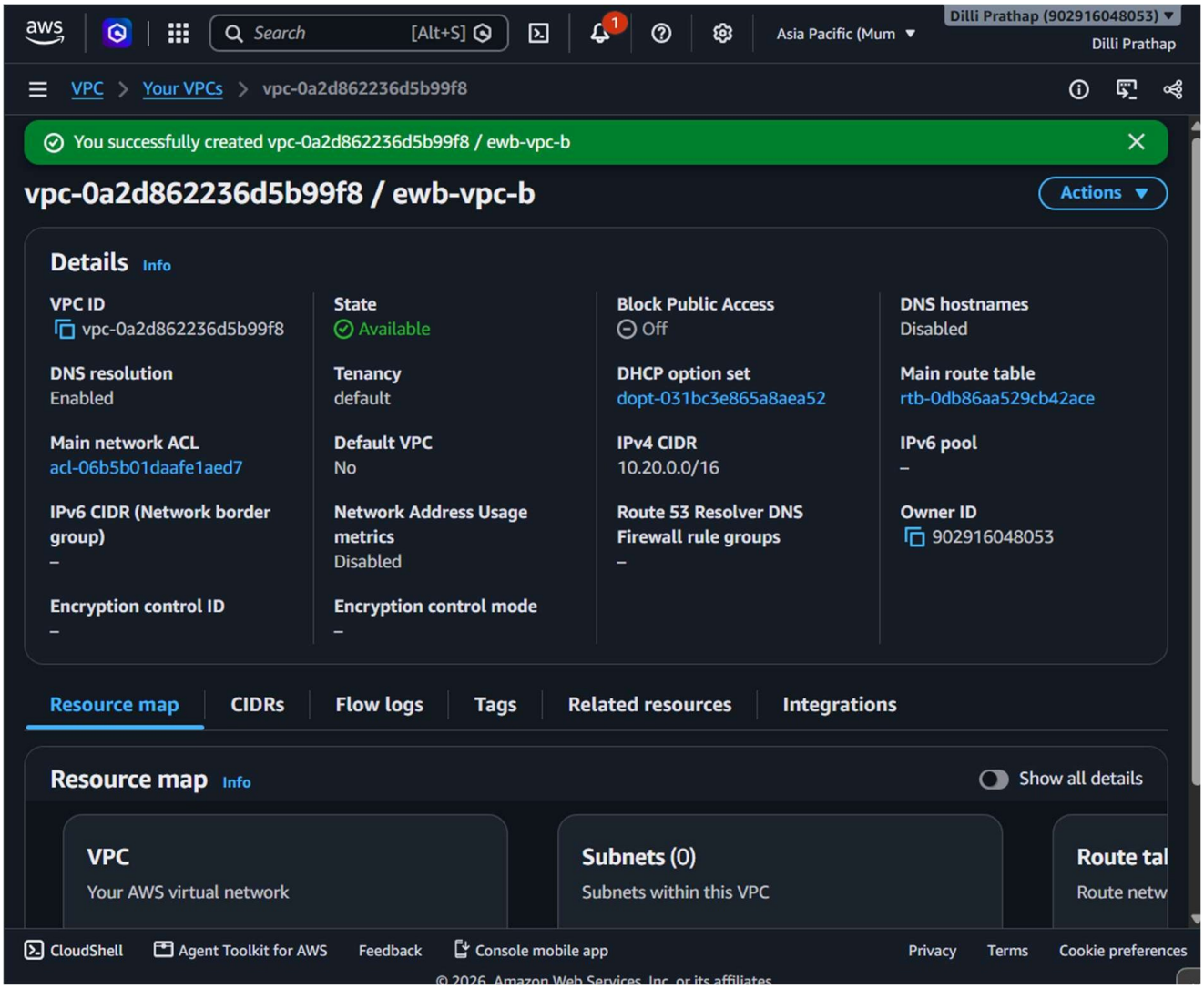
Host / console page	Test / checkpoint	Result
Route tables > ewb-rt-a-private	Association, no internet route	Associated with ewb-a-private. Only route: 10.10.0.0/16 local (no 0.0.0.0/0)



4. Build evidence - VPC B and peering

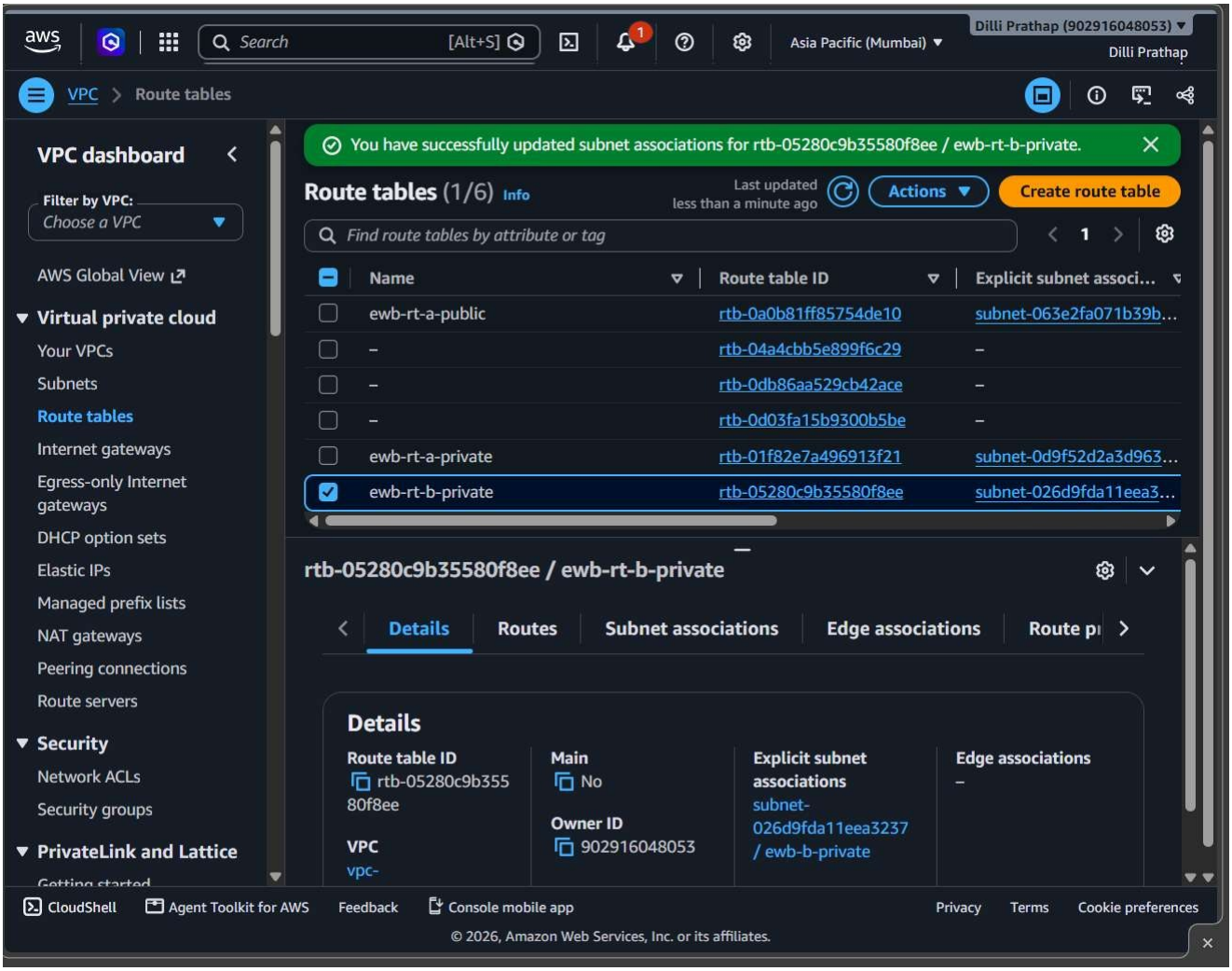
4.1 VPC B created (ewb-vpc-b)

Host / console page	Test / checkpoint	Result
VPC console > Your VPCs	Create VPC	State Available; IPv4 CIDR 10.20.0.0/16; ID vpc-0a2d862236d5b99f8; does not overlap VPC A



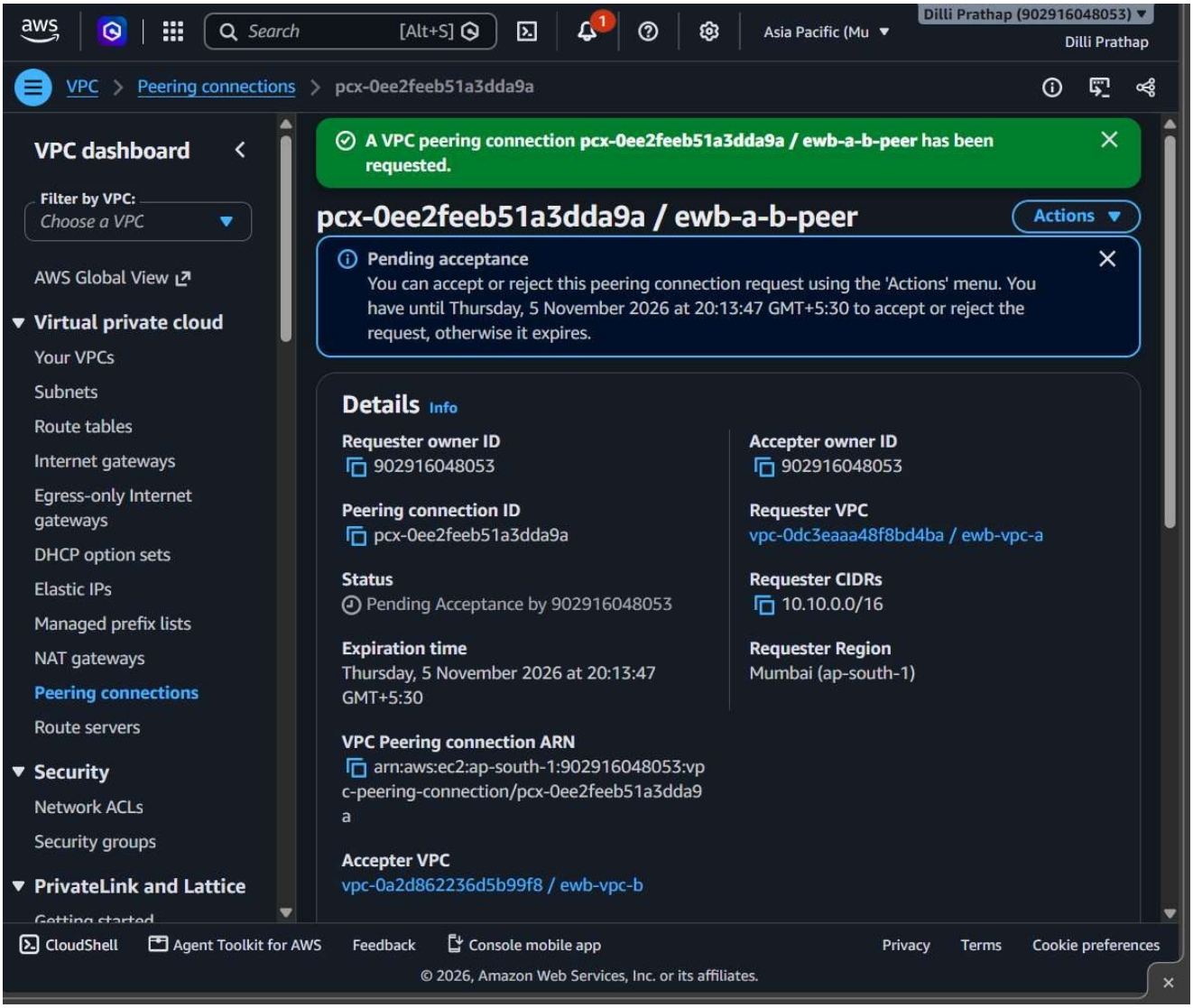
4.2 ewb-rt-b-private associated with ewb-b-private

Host / console page	Test / checkpoint	Result
VPC console > Route tables	VPC B route table	rtb-05280c9b35580f8ee / ewb-rt-b-private associated with subnet-026d9fda11eea3237 / ewb-b-private



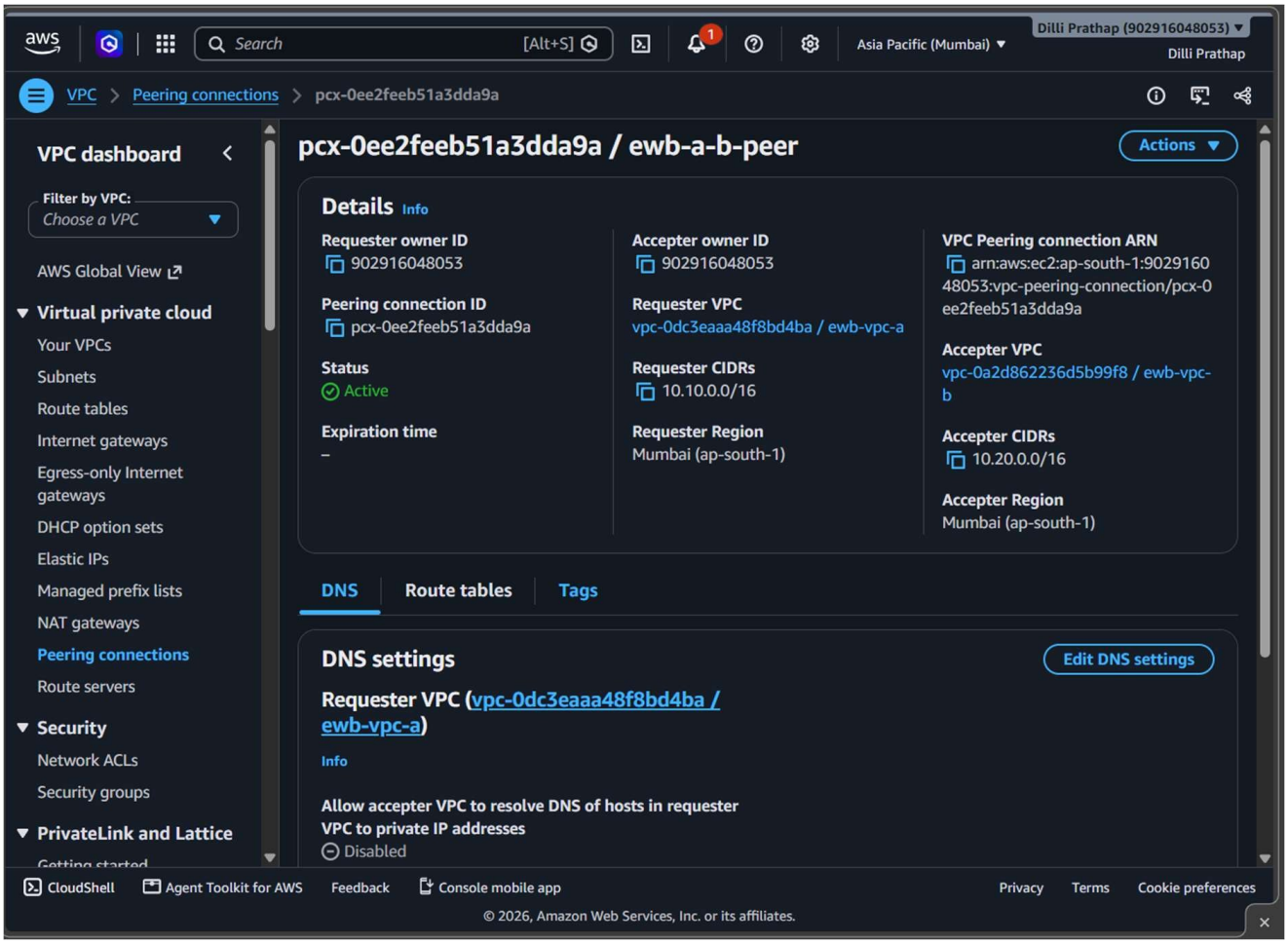
4.3 Peering connection requested

Host / console page	Test / checkpoint	Result
VPC console > Peering connections	Create peering request	pcx-0ee2feeb51a3dda9a / ewb-a-b-peer: requester vpc-0dc3eaaa48f8bd4ba (10.10.0.0/16), accepter vpc-0a2d862236d5b99f8



4.4 Peering connection Active

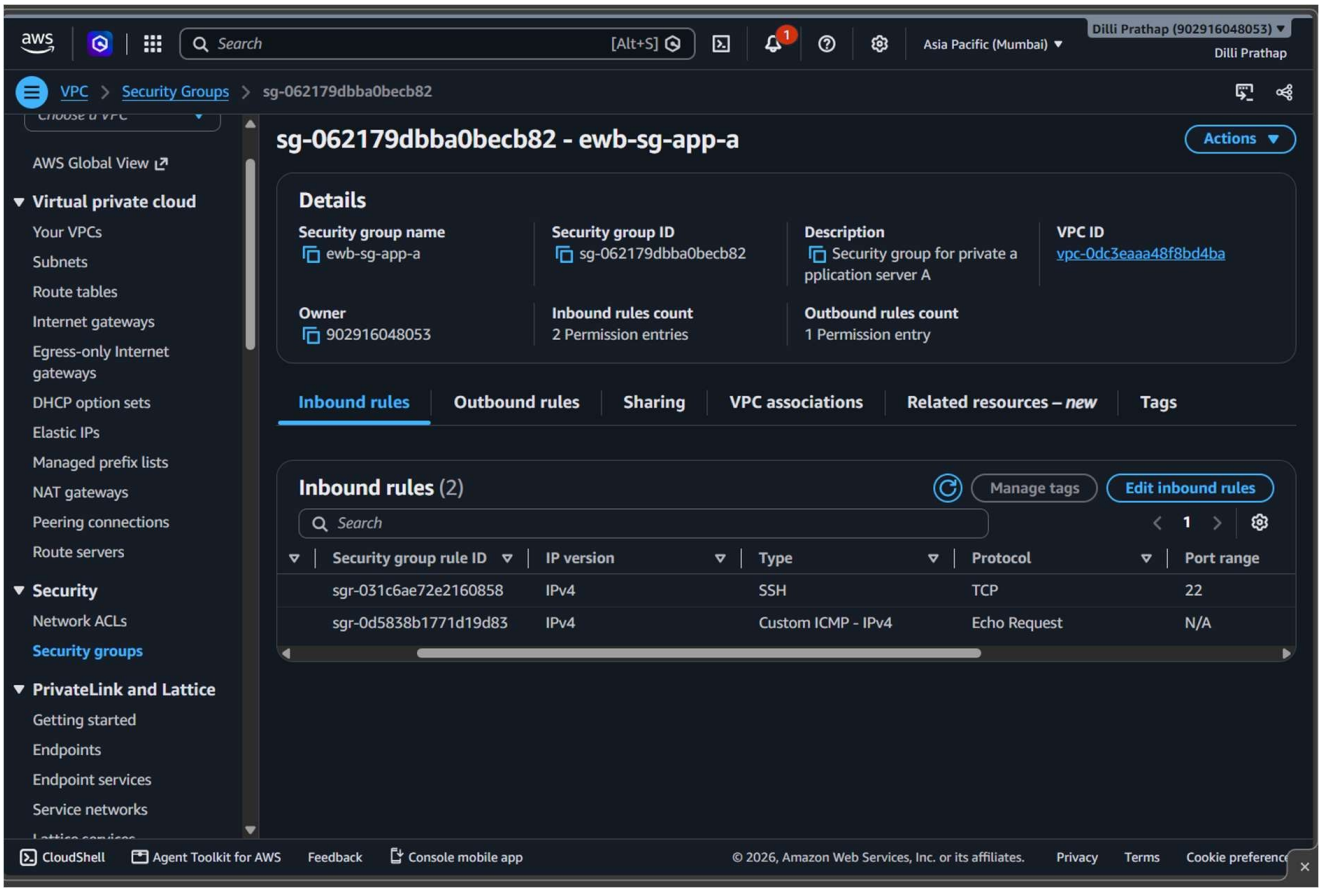
Host / console page	Test / checkpoint	Result
VPC console > Peering connections	Accept request	Status Active. Requester 10.10.0.0/16 and accepter 10.20.0.0/16, both in Mumbai (ap-south-1)



5. Security groups and instances

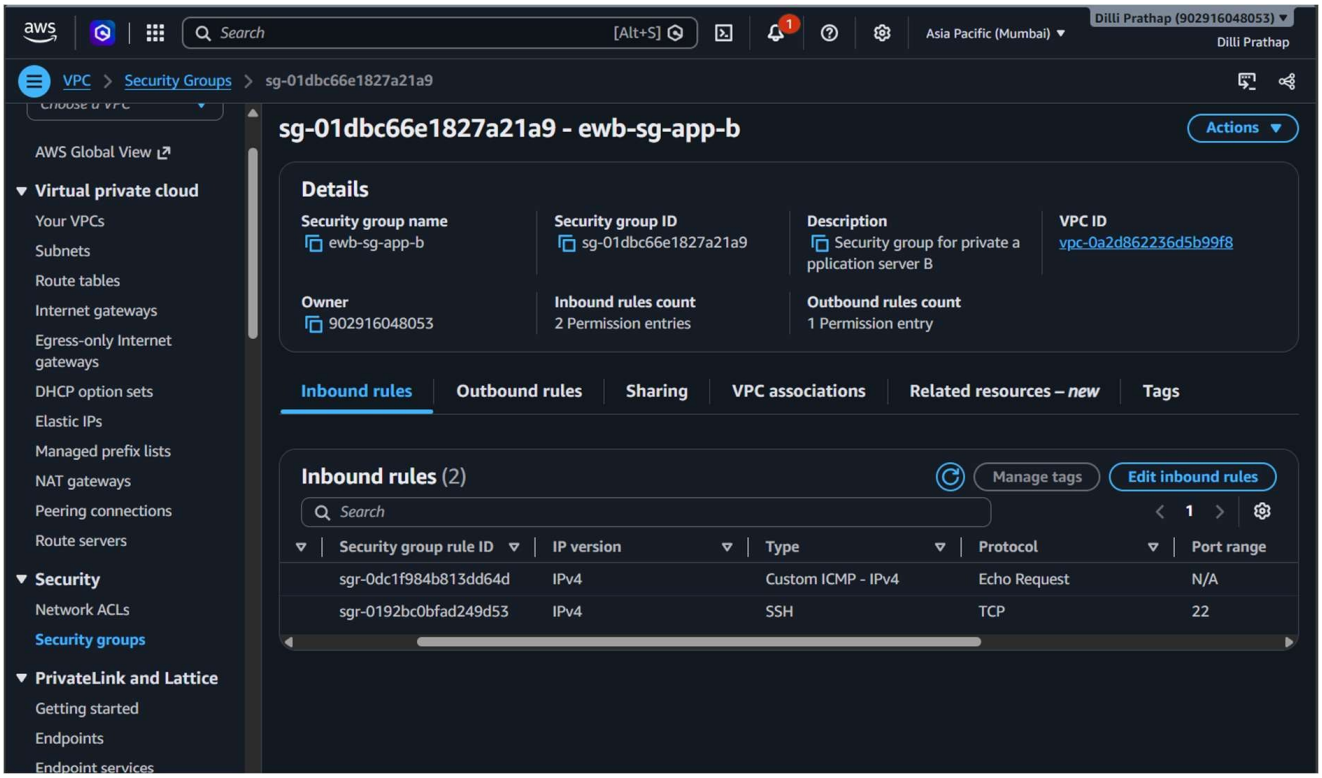
5.1 Security group ewb-sg-app-a

Host / console page	Test / checkpoint	Result
VPC console > Security groups	Inbound rules	Inbound rules (2): SSH TCP 22 and Custom ICMP IPv4 Echo Request. VPC vpc-0dc3eaaa48f8bd4ba



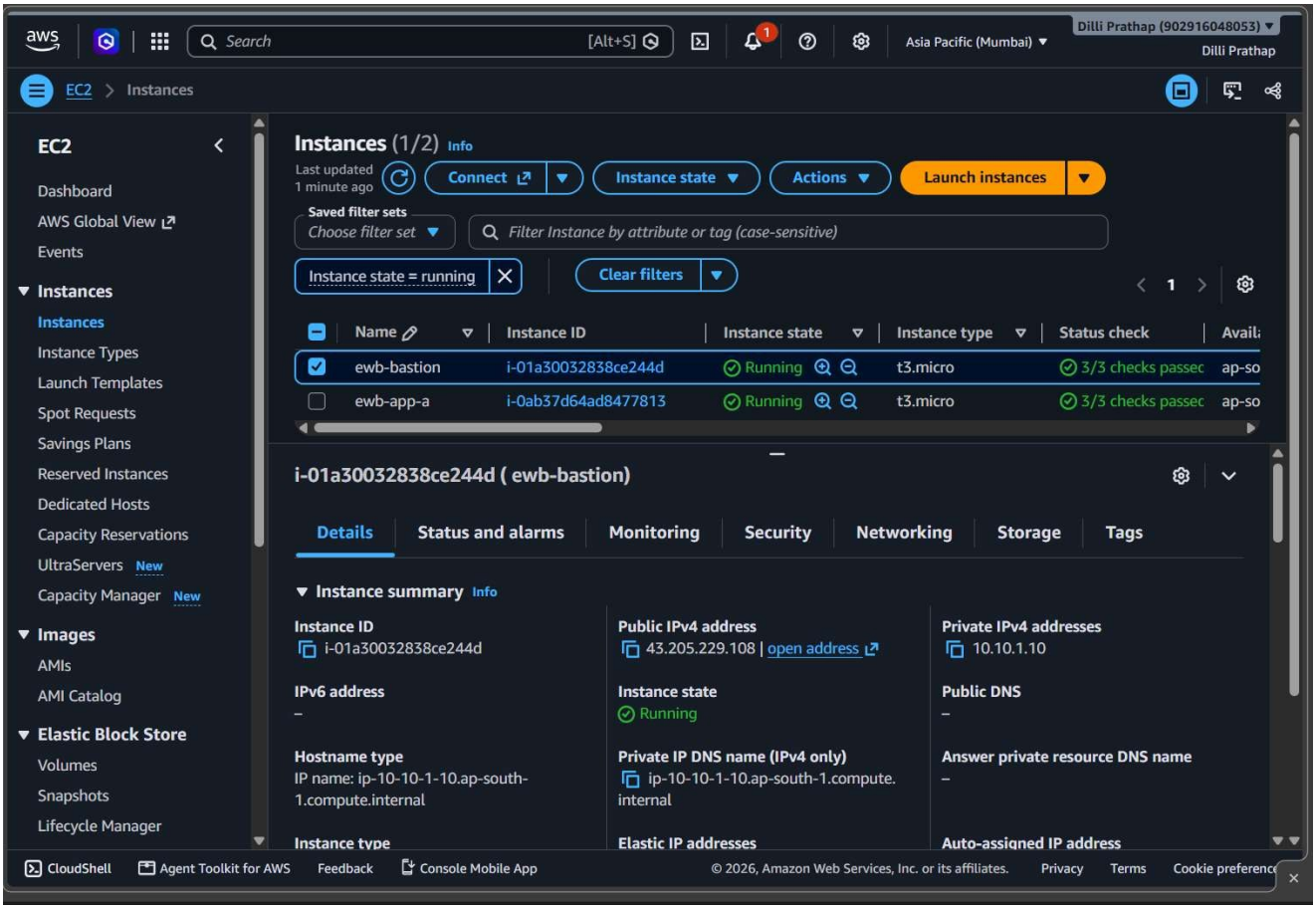
5.2 Security group ewb-sg-app-b

Host / console page	Test / checkpoint	Result
VPC console > Security groups	Inbound rules	Inbound rules (2): Custom ICMP IPv4 Echo Request and SSH TCP 22. VPC vpc-0a2d862236d5b99f8



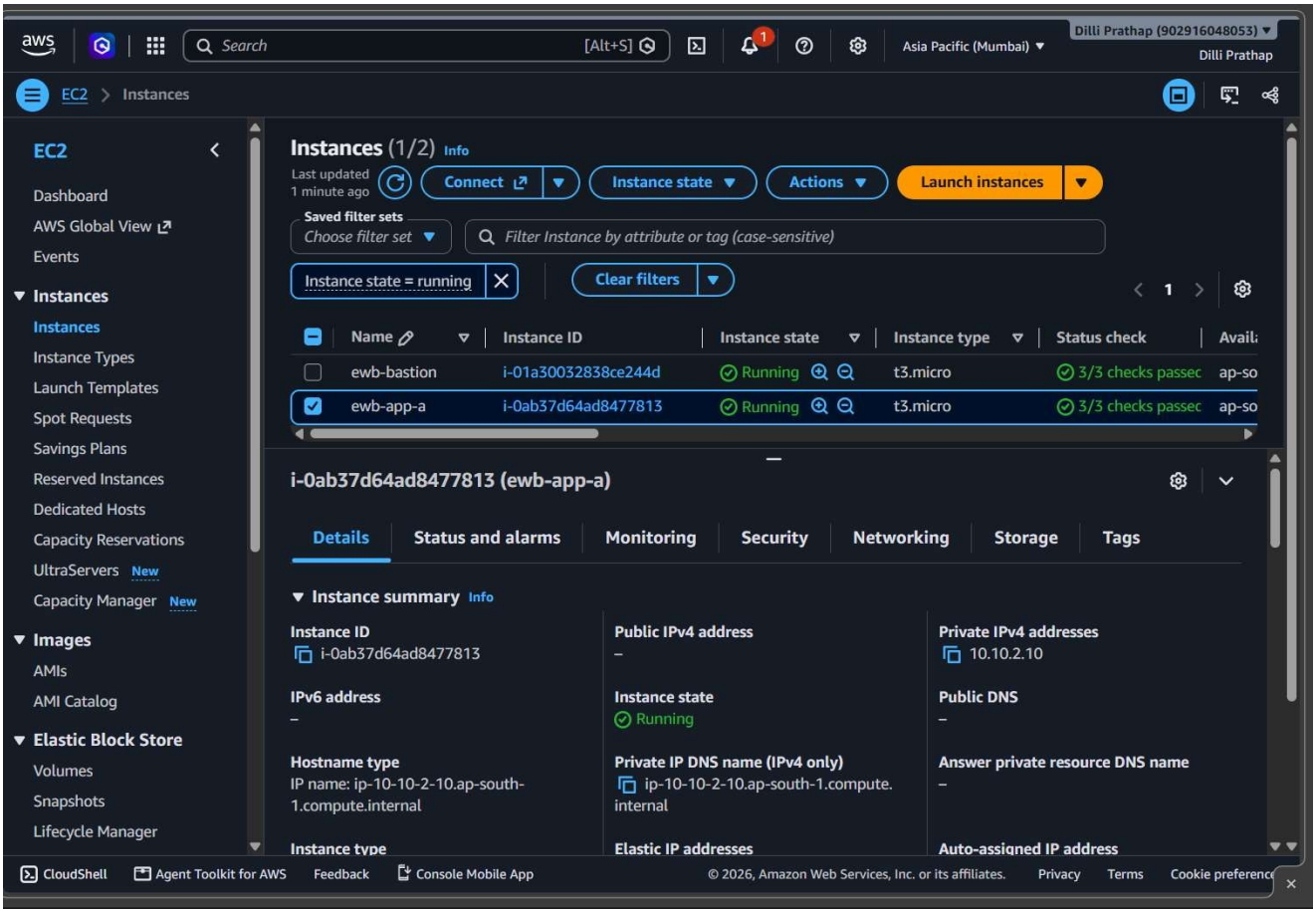
5.3 Bastion host running

Host / console page	Test / checkpoint	Result
EC2 console > Instances	Bastion instance	ewb-bastion i-01a30032838ce244d Running, 3/3 checks passed; public IPv4 43.205.229.108; private IP 10.10.1.10



5.4 Application server A running with no public IP

Host / console page	Test / checkpoint	Result
EC2 console > Instances	App-A instance	ewb-app-a i-0ab37d64ad8477813 Running, 3/3 checks passed; Public IPv4 address empty; private IP 10.10.2.10

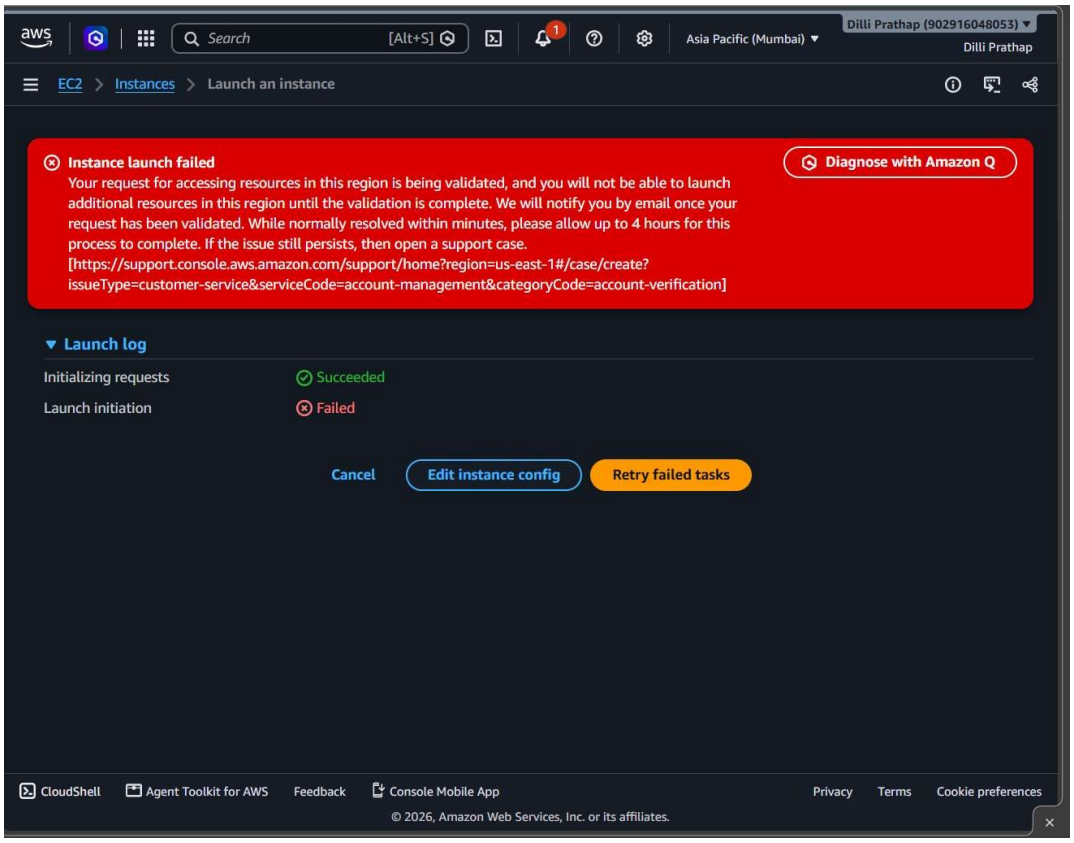


6. Note on application server B

When I launched ewb-app-b, AWS blocked the launch with an account validation message for the region (shown below). This is an account-level restriction applied by AWS to new launches in the region, not a configuration error in my VPC, subnet or security group. AWS states it normally resolves within minutes, and up to 4 hours at most, after which the launch can be retried.

Instance launch message from AWS

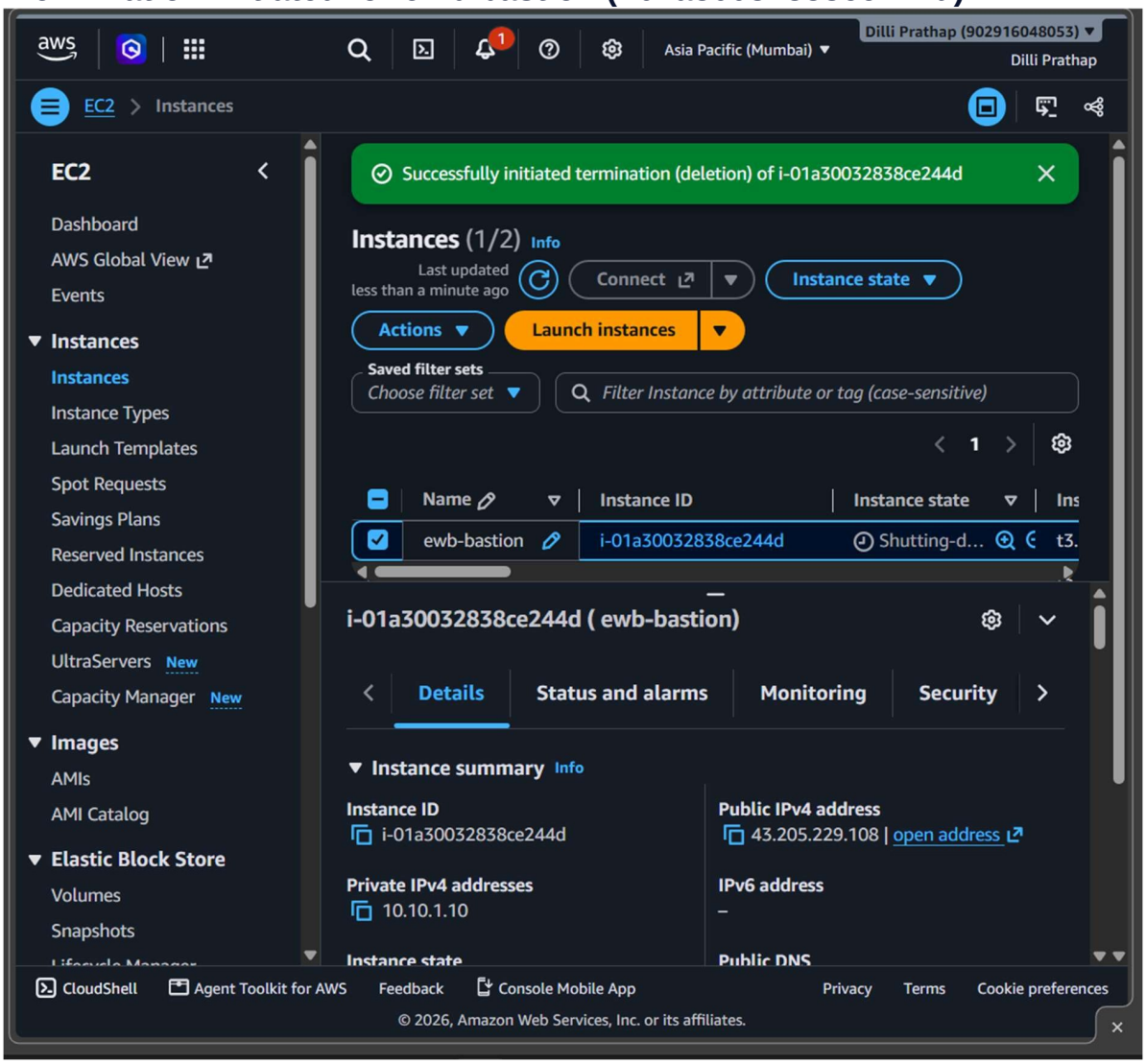
Host / console page	Test / checkpoint	Result
EC2 console > Launch an instance	Launch App-B	Request validation for the region in progress; launch initiation blocked by AWS



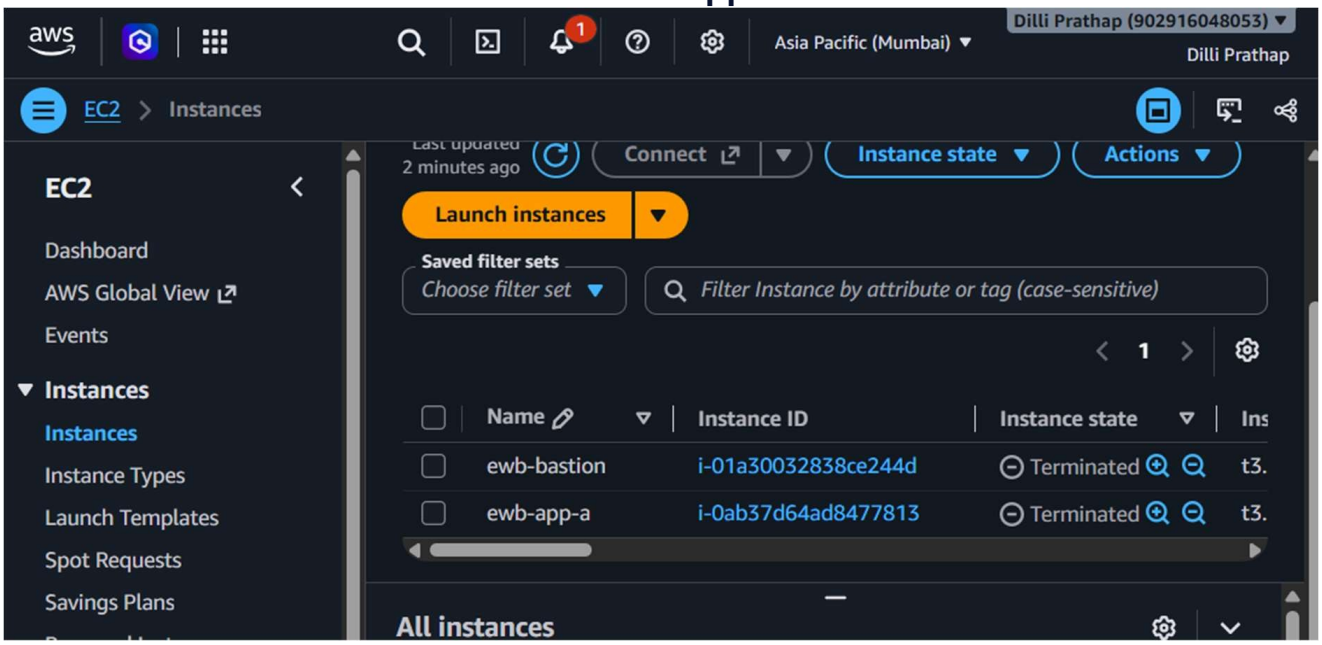
7. Cleanup and termination

After testing, I removed the resources so no charges continue: instances first, then security groups, the peering connection, subnets, route tables, and finally detached the internet gateway.

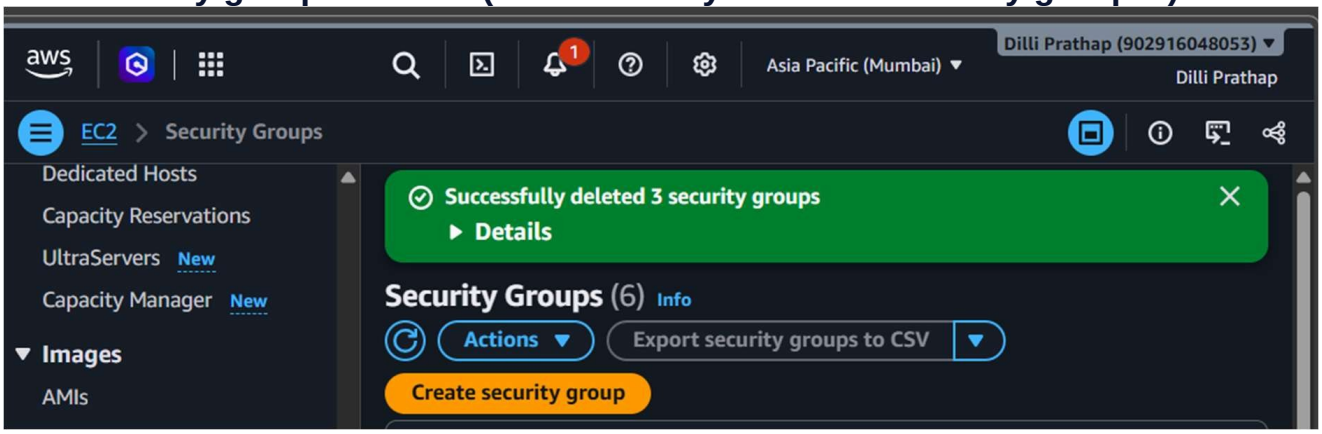
7.1 Termination initiated for ewb-bastion (i-01a30032838ce244d)



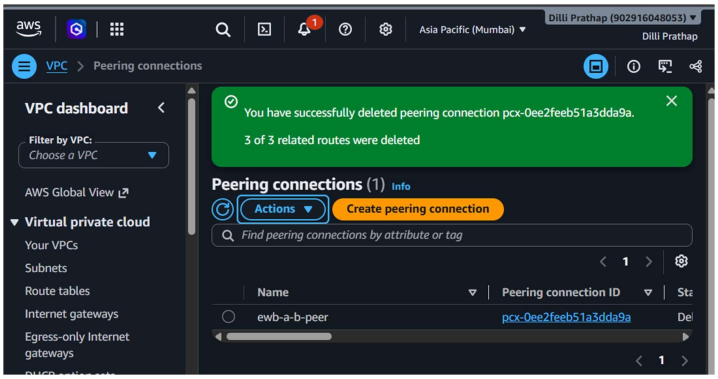
7.2 EC2 instances ewb-bastion and ewb-app-a show Terminated



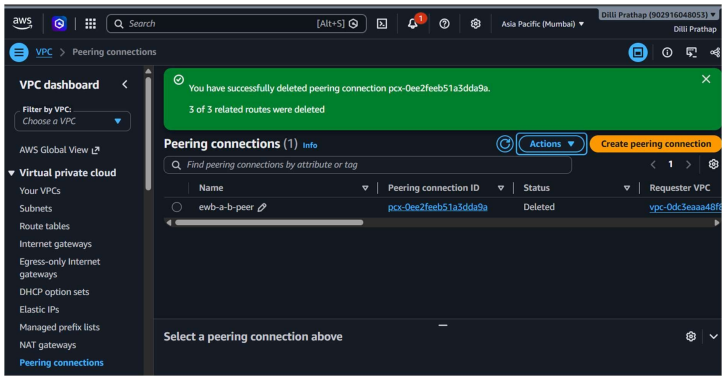
7.3 Security groups deleted ("Successfully deleted 3 security groups")



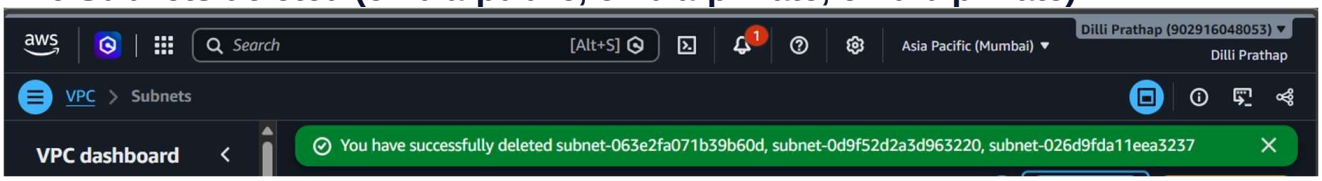
7.4 Peering connection pcx-0ee2feeb51a3dda9a deleted (3 of 3 related routes deleted)



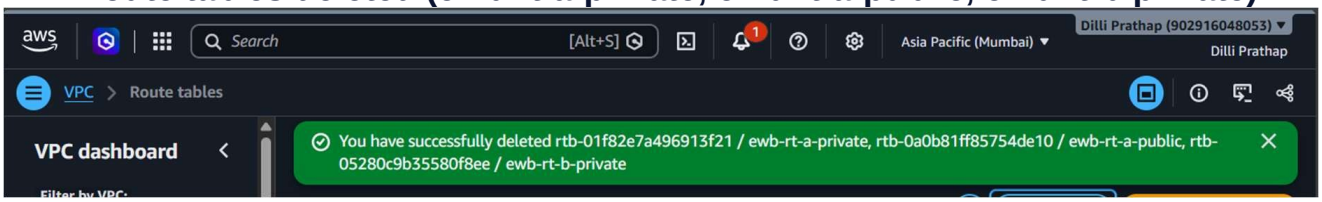
7.5 ewb-a-b-peer status Deleted



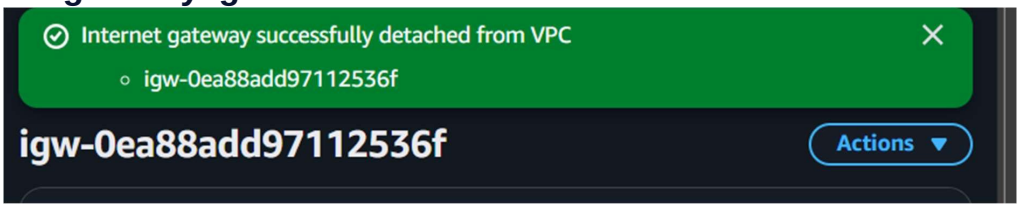
7.6 Subnets deleted (ewb-a-public, ewb-a-private, ewb-b-private)



7.7 Route tables deleted (ewb-rt-a-private, ewb-rt-a-public, ewb-rt-b-private)



7.8 Internet gateway igw-0ea88add97112536f detached from the VPC



8. Design explanation: forward path and return path

Why each route and rule is needed

Item	Why it is needed
Local routes	Every route table gets the VPC CIDR as 'local'. This is how hosts inside one VPC reach each other.
ewb-rt-a-private: 10.20.0.0/16 to pcx (forward path)	Packets from App-A to 10.20.1.10 would otherwise match no route and be dropped. This sends them into the peering connection.
ewb-rt-b-private: 10.10.0.0/16 to pcx (return path)	App-B must know how to reply to 10.10.2.10. A one-way route is not enough; replies need a route back too.
ewb-rt-a-public: 10.20.0.0/16 to pcx	With ProxyJump, the bastion itself opens the SSH connection to App-B, so its subnet needs a route to VPC B. The more specific /16 wins over 0.0.0.0/0.
Active peering is not enough	Peering is only the link. Route tables and security groups decide whether traffic may use it.
SSH source on App-B = bastion private IP	App-B sees the bastion's private address 10.10.1.10 as the source, not my laptop's public IP.
ICMP Echo Request rules	Each app server accepts ping only from the other app server. Security groups are stateful, so replies are allowed automatically.
No internet gateway on VPC B	Peering is non-transitive and carries only traffic between the two VPC CIDRs, so VPC B cannot use VPC A's internet gateway.

Route design for the peering

Route table	Destination	Target
ewb-rt-a-private	10.10.0.0/16 / 10.20.0.0/16	local / pcx-0ee2feeb51a3dda9a
ewb-rt-b-private	10.20.0.0/16 / 10.10.0.0/16	local / pcx-0ee2feeb51a3dda9a
ewb-rt-a-public	10.10.0.0/16 / 0.0.0.0/0 / 10.20.0.0/16	local / igw-0ea88add97112536f / pcx-0ee2feeb51a3dda9a